

Towards Safer Mixed Reality with Auditory Warnings for Uncontrolled Indoor Settings

Nasim Ahmed *
Computer Science

Taeyeong Choi†
Information Technology

Xinyue Zhang‡
Computer Science

Sahidul Islam §
Computer Science

Rifatul Islam¶
Computer Science

Kennesaw State University, GA, USA

ABSTRACT

Mixed Reality (MR) applications are increasingly gaining popularity across various sectors, making user safety crucial, as distractions from augmented and virtual content can cause injuries. Native safety features such as user-defined Guardian boundaries offer limited protection in controlled indoor settings. While limited research has been done to address safety in dynamic uncontrolled environments, we propose a novel method providing verbal feedback to help users safely navigate uncontrolled indoor settings. Our proposed approach achieved a mean accuracy of **93.64%** with an average latency of **380 ms**; thus demonstrating promising results for ensuring safety in MR interactions.

Index Terms: Mixed Reality, User Safety, Hazard Detection, and Auditory Feedback.

1 INTRODUCTION

Mixed Reality (MR) enables virtual and real objects to coexist and interact seamlessly [6]. However, interacting with augmented virtual objects can easily distract users, potentially leading to hazardous situations and increasing the risk of injuries to both users and bystanders [4]. Native safety features, such as Meta's Guardian Boundary system, provide protection in user-defined controlled indoor environments but are limited by their inability to adapt to uncontrolled settings, such as free navigation in indoor and outdoor spaces, tracking moving objects, detecting dynamic hazards, or preventing collisions with bystanders. Smartphone-based augmented reality (AR) applications are available for detecting potential hazards in front of users; however, they require holding the device to a certain orientation [3, 8]. Additionally, while some research has investigated delivering real-time hazard warnings on construction sites through extensive CCTV video data collection, this method is impractical for consumer-grade MR devices (e.g., Meta Quest, Apple Vision Pro) and everyday scenarios [10]. Prior research also introduced head-mounted display (HMD) prototypes capable of facilitating transitions between real and virtual environments, but these systems lack interactivity and fail to map real-world surroundings or provide responsive feedback essential for user safety [1, 2]. As MR systems are becoming more popular in diverse use cases in indoor and outdoor uncontrolled environments, ensuring users' safety becomes one of the critical research challenges. To address this research gap, we present our preliminary findings on a closed-loop MR system that uses on-device integrated cameras to detect potential hazards in uncontrolled indoor environments and provide users with auditory

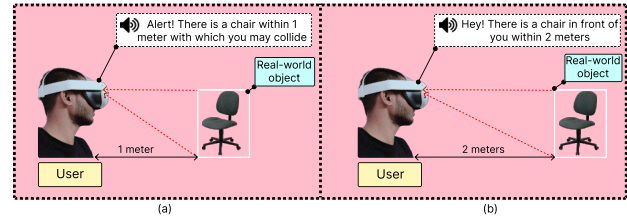


Figure 1: HMD-based Mixed Reality application with verbal feedback for safe walking in indoor conditions: (a) Warning for potential collision (b) Warning for potential hazard.

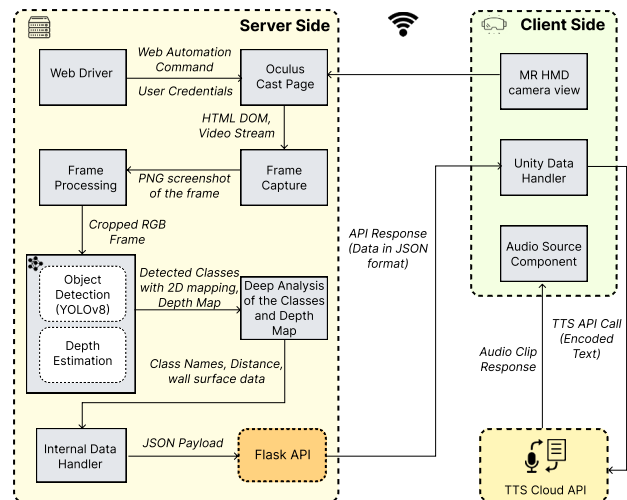


Figure 2: System Architecture and Workflow of the Application combining all components and principal modules

warnings. Our proposed system achieves a mean accuracy of 93.64% with an average latency of 380 ms.

A conceptual visualization of the application is shown in Figure 1. Our application was primarily developed and deployed to **Meta Quest 3**, a standalone MR headset by Meta. The proposed solution demonstrated robust performance in several experimental trials and provided real-time hazard detection and localization.

2 SYSTEM DESIGN

The application was designed for a mixed reality environment, and its overall flow and architecture are shown in Figure 2. It comprises three broader modules: the server side, the client side (i.e. user side), and the Text-to-Speech (TTS) Cloud API. The system was designed wireless, and the data transmission among the modules happened under the HTTP protocol.

*e-mail: nahmed25@students.kennesaw.edu

†e-mail: tchoi3@kennesaw.edu

‡e-mail: xzhang48@kennesaw.edu

§e-mail: sislam19@kennesaw.edu

¶e-mail: rislam11@kennesaw.edu

2.1 Server Side

The server-side component processes video frames, detects objects, and estimates depth in real-time. Video frames for the HMD camera view are captured from the Oculus cast page by interacting with the HTML DOM (Document Object Model) through a Chromium web driver. These video frames are then processed by OpenCV to extract the relevant portions of the MR environment. The extracted frames are then passed to the computer vision module, where the processed RGB frames will go through two computer vision models: YOLOv8 [9], and Depth estimation model (midas_v21_small_256) [5] for further processing. These pre-trained models were chosen for their enhanced performance and robustness compared to their other alternatives in their domain. The object detection model identifies objects, assigns bounding boxes, and outputs class names with confidence scores. A particular object is processed further if its confidence score exceeds 50%. Meanwhile, the pre-trained depth estimation model generates a depth map from RGB frames for two purposes. First, it detects walls by applying a depth threshold (1000mm or 1 meter) and contour analysis. This thresholding process generates a binary mask, which is then applied to the contour detection. By filtering contours based on their area and shape, flat and continuous surfaces corresponding to walls are identified. Second, the depth map is utilized to measure the distance between the user and the central point of the detected object, focusing on the most stable point to minimize errors. After having all of the data about the walls, detected objects, and distance, a Flask API handles communication between the server and the HMD, transmitting the data to the client side in JSON format.

2.2 Client Side

On the client side, the system operates through a unity application installed in the **Meta Quest 3**. The application runs in passthrough mode on the headset (disabling the guardian system) and captures the real-time view of the surroundings to stream to the Oculus casting page, from which the server can receive and process it. Once data is obtained from the server, unity processes them with key constraints such as which objects to consider first, whether it is a potential hazard, whether there is a wall in front, etc. The potential danger zone was set to 1 meter in front of the user, which means if the object is within a 1-meter distance in front, then the application will alert the user. After processing all those things, it sets a text in string format which is provided to the user as verbal feedback. For example, if a chair is within a 1-meter distance, it sets the string as "Alert! There is a chair within 1 meter with which you may collide." If no objects are detected within the immediate danger zone, but potential obstacles are present within the hazard zone (extending up to 2 meters in front of the user), the application prioritizes the nearest 1 to 3 objects for feedback. This approach prevents hearing discomfort and minimizes cognitive load.

2.3 Text-to-Speech (TTS) Cloud API

The TTS API converts string alerts from the Unity application into auditory alerts, enhancing situational awareness and navigation safety. After receiving object and depth data from the Flask API, Unity formats and sends this data to the StreamElements TTS API [7], which then processes and returns it as verbal feedback in audio format. This ensures users receive real-time auditory alerts.

3 RESULTS

For evaluation of diversity and acknowledging the necessity of adequate lights, we assessed the performance of the application with three different indoor conditions: The **well-lit Environment** (Condition A), **Dimly-lit Environment** (Condition B), and **Cluttered Environment** (Condition C). We tested our application with 15 trials (5 trials for each condition) with different indoor components such as chairs, tables, persons, walls, etc. The application's performance for

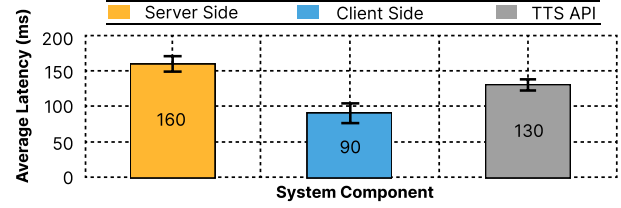


Figure 3: Latency analysis across principal system components

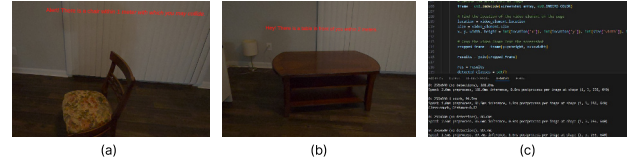


Figure 4: Result visualization, and validation using textual overlays. (a) An object is within a 1-meter distance, (b) An object is within a 2-meter distance (c) Object detection with depth data on the server side

different conditions is shown in the table 1. The detection accuracy was measured by dividing the number of true detections by the total detections for each trial, and then averaging the results across all trials for each condition. The experiment was conducted with several objects (11 in condition 1, 9 in condition 2, and 7 in condition 3) for each condition. As shown in Table 1, the application achieved remarkable accuracy across different indoor conditions with an overall mean accuracy of **93.64%**. Figure 4 shows the visualization of the textual overlays of the verbal feedback in a test version of the application (a, and b), and the object detection with depth data on the server side (c). To avoid distraction, the overlays were not present in the original application. Figure 3 shows the latency for each principal system component on a 5GHz Wi-Fi (802.11ac) network, highlighting the need for server-side improvements.

Table 1: Detection accuracy and depth estimation error under different indoor conditions.

Condition	Mean Detection Accuracy (%)	Depth Estimation Error (Mean $\pm$ SD) (cm)
Condition A (Well-lit)	96.18	4.3 $\pm$ 1.1
Condition B (Dimly-lit)	93.33	6.1 $\pm$ 1.8
Condition C (Cluttered)	91.42	7.6 $\pm$ 2.0
Overall Mean	93.64	6.0 $\pm$ 1.6

4 CONCLUSION & FUTURE WORKS

Our system demonstrated strong performance and met initial objectives. However, some limitations require attention. Improvements are needed in detection accuracy and distance estimation precision. Acknowledging the device dependency, we plan to expand it to a broader version with multiple features included while making it ubiquitous in any MR device. For this, we plan to use an external RGBD camera to overcome the issue of depth mapping and camera access. In the future version, we are also thinking about the privacy concerns of the users and the bystanders, and we aim to implement mechanisms that anonymize sensitive visual data before it is transmitted to the server for processing while maintaining the performance of the proposed models.

REFERENCES

- [1] J. Jung, H. Lee, J. Choi, A. Nanda, U. Gruenefeld, T. Stratmann, and W. Heuten. Ensuring safety in augmented reality from trade-off between immersion and situation awareness. In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 70–79, 2018. doi: 10.1109/ISMAR.2018.00032 [1](#)
- [2] K. Kanamori, N. Sakata, T. Tominaga, Y. Hijikata, K. Harada, and K. Kiyokawa. Obstacle avoidance method in real space for virtual reality immersion. In *2018 IEEE International Symposium on Mixed and Augmented Reality (ISMAR)*, pp. 80–89, 2018. doi: 10.1109/ISMAR.2018.00033 [1](#)
- [3] H. Kang, G. Lee, and J. Han. Obstacle detection and alert system for smartphone ar users. In *Proceedings of the 25th ACM Symposium on Virtual Reality Software and Technology, VRST '19*. ACM, 2019. [1](#)
- [4] T. Papadopoulos, K. Evangelidis, T. Kaskalis, G. Evangelidis, and S. Sylaiou. Interactions in augmented and mixed reality: An overview. *Applied Sciences*, 11:8752, 09 2021. doi: 10.3390/app11188752 [1](#)
- [5] R. Ranftl, K. Lasinger, D. Hafner, K. Schindler, and V. Koltun. Towards robust monocular depth estimation: Mixing datasets for zero-shot cross-dataset transfer, 2020. [2](#)
- [6] S. Schmidt, O. J. A. Nunez, and F. Steinicke. Blended agents: Manipulation of physical objects within mixed reality environments and beyond. In *Symposium on Spatial User Interaction, SUI '19*. ACM, 2019. [1](#)
- [7] StreamElements. Setting up command-triggered text-to-speech (tts). <https://support.streamelements.com>. Accessed: 2024-12-23. [2](#)
- [8] C. H. Tsai, F. Elyasi, P. Ren, and R. Manduchi. All the way there and back: Inertial-based, phone-in-pocket indoor wayfinding and backtracking apps for blind travelers. *ACM Trans. Access. Comput.*, 17(4), Dec. 2024. doi: 10.1145/3696005 [1](#)
- [9] R. Varghese and S. M. Yolov8: A novel object detection algorithm with enhanced performance and robustness. In *ADICS 2024*, pp. 1–6, 2024. doi: 10.1109/ADICS58448.2024.10533619 [2](#)
- [10] S. Wu, L. Hou, G. K. Zhang, and H. Chen. Real-time mixed reality-based visual warning for construction workforce safety. *Automation in Construction*, 139:104252, 2022. [1](#)